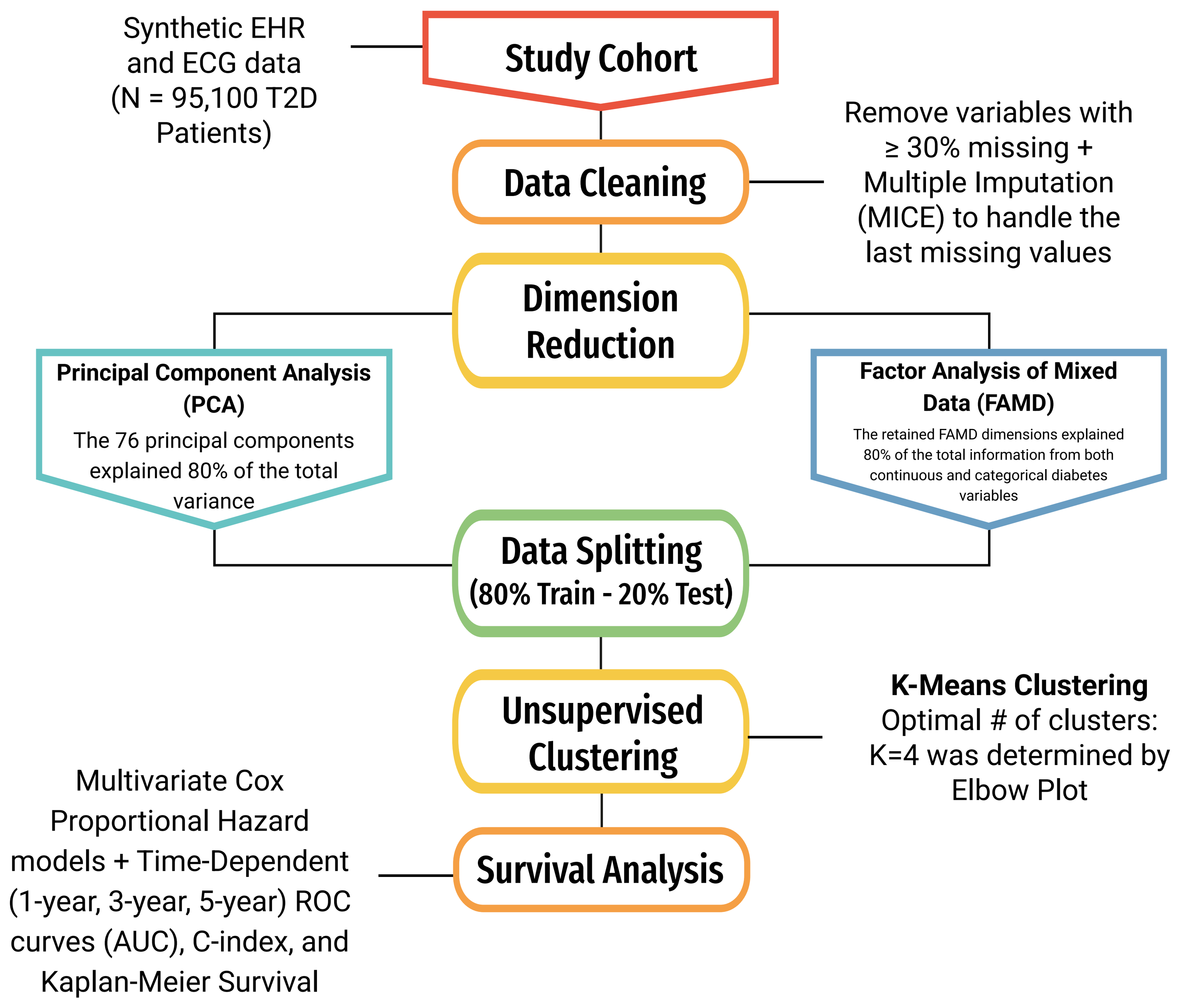
Identifying Cardiovascular Risk Phenotypes in Type 2 Diabetes: A Clustering Approach Using Electronic Health Record and ECG Data

Xinyi Sheng*, Shirling Liu, Yutian Wang, Wendy Shue (Advisor: Dr. Wendy Lou)
Dalla Lana School of Public Health, University of Toronto, Toronto, ON; *Team captain

Introduction

- **Type 2 diabetes (T2D)** accounts for 90–95% of diabetes cases, confers a 2- to 4-fold increased risk of major adverse cardiovascular events (MACE), and ~25% of U.S. diabetes cases remain undiagnosed.[1,2]
- MACE is a leading cause of morbidity and mortality in T2D, yet current prediction models treat T2D as homogeneous, rely on traditional variables with only modest performance, and often require data not routinely available in clinical practice.[3,4]
- Routine clinical care generates vast underutilized Electronic Health Record (EHR) and 12-lead Electrocardiogram (ECG) data; unsupervised clustering of these data can uncover hidden phenotypes, improving risk stratification beyond traditional trials and advancing precision cardiovascular risk prediction.[1,3,5]
- This study aims to identify clinically meaningful phenotypes among patients with T2D using integrated EHR and ECG variables and evaluate their value for improving MACE risk prediction and supporting personalized diabetes care.

Methods


Synthetic EHR and ECG data (N = 95,100 T2D Patients)
Study Cohort
Data Cleaning
Remove variables with ≥ 30% missing + Multiple Imputation (MICE) to handle the last missing values
Dimension Reduction
Principal Component Analysis (PCA)
The 76 principal components explained 80% of the total variance
Factor Analysis of Mixed Data (FAMD)
The retained FAMD dimensions explained 80% of the total information from both continuous and categorical diabetes variables
Data Splitting
(80% Train - 20% Test)
Unsupervised Clustering
K-Means Clustering
Optimal # of clusters: K=4 was determined by Elbow Plot
Survival Analysis
Multivariate Cox Proportional Hazard models + Time-Dependent (1-year, 3-year, 5-year) ROC curves (AUC), C-index, and Kaplan-Meier Survival

Results

PCA

- **Phenotype Discovery:** PCA captured 80% of total variance via 76 components, revealing 4 distinct T2D phenotypes through unsupervised clustering.
- **Predictive Accuracy:** The cluster-enhanced Cox model achieves strong discriminative performance (Test AUCs: 0.772 for 1-year, 0.777 for 3-year, 0.783 for 5-year), consistent with training results (Table 3).
- **Prognostic Significance:** KM analysis (Figure 1 and Table 1) revealed significant risk divergence across phenotypes ($p < 0.0001$; 10.9%–79.6% MACE). Cluster membership significantly improved predictive power over baseline demographics (Likelihood Ratio test, $p < 2.2e^{-16}$).

FAMD

- **Phenotype Discovery:** FAMD captured 80% of the total information from both continuous and categorical variables, revealing 4 distinct T2D phenotypes through unsupervised clustering.
- **Predictive Accuracy:** According to Table 3, the cluster-enhanced Cox model showed strong predictive performance for MACE risk across 1-year (Test AUC: 0.771), 3-year (Test AUC: 0.777), and 5-year (Test AUC: 0.777) outcomes.
- **Prognostic Significance:** KM analysis (Figure 2) showed clear survival differences across FAMD-derived phenotypes ($p < 0.0001$; 10.87%–79.29% MACE), suggesting that cluster membership was associated with different levels of cardiovascular risk (Cox Likelihood Ratio test, $p < 2.2e^{-16}$).

Table 1. Cluster Profile (PCA Training Set)

Cluster	N	Percent	Mean age	Female %	Male %	Overall MACE	1y MACE	3y MACE	5y MACE
1	5,976	7.9%	72.4	32.1%	67.9%	79.6%	26.0%	38.0%	47.5%
2	10,228	13.4%	64.0	11.4%	88.6%	43.0%	7.7%	14.5%	22.3%
3	27,625	36.3%	68.1	44.2%	55.8%	65.7%	16.9%	28.6%	39.3%
4	32,251	42.4%	53.0	48.7%	51.3%	10.9%	1.9%	3.8%	5.8%

Table 2. Cluster Profile (FAMD Training Set)

Cluster	N	Percent	Mean age	Female %	Male %	Overall MACE	1y MACE	3y MACE	5y MACE
1	5,982	7.86%	72.39	31.98%	68.02%	79.29%	25.93%	38.30%	49.30%
2	10,340	13.59%	64.00	11.49%	88.51%	43.18%	7.77%	14.87%	24.73%
3	27,526	36.18%	68.11	44.32%	55.68%	65.75%	19.96%	29.17%	41.72%
4	32,232	42.37%	52.84	48.64%	51.36%	10.87%	1.85%	3.90%	6.18%

Figure 1. Kaplan-Meier Survival Curve by Cluster (Testing set)

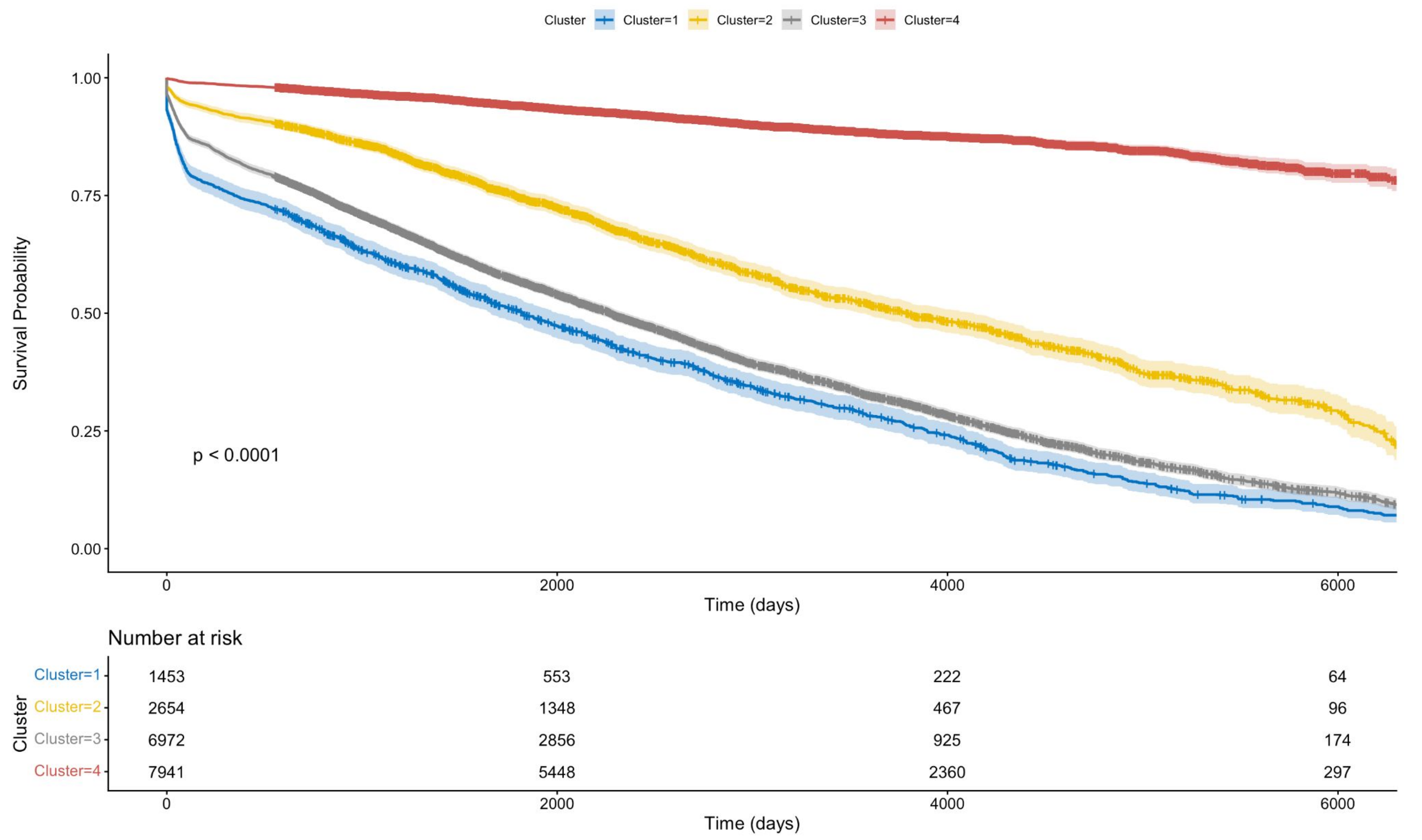

Cluster
Cluster=1
Cluster=2
Cluster=3
Cluster=4
Survival Probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
Time (days)
Number at risk
Cluster
Cluster=1 1453 553 222 64
Cluster=2 2654 1348 467 96
Cluster=3 6972 2856 925 174
Cluster=4 7941 5448 2360 297
Time (days)

Figure 2. Kaplan-Meier Survival Curve by Cluster (Testing set)

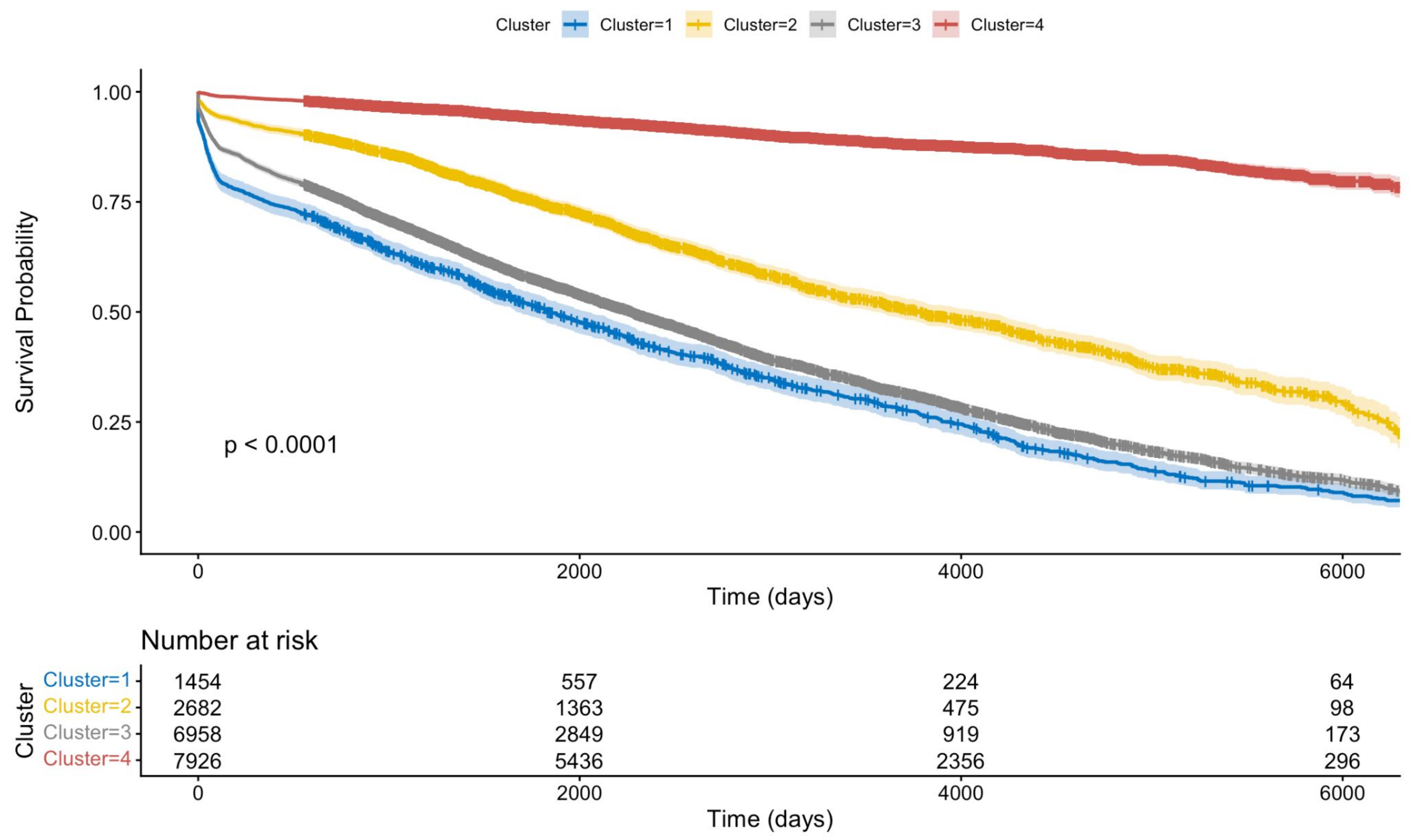

Cluster
Cluster=1
Cluster=2
Cluster=3
Cluster=4
Survival Probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
Time (days)
Number at risk
Cluster
Cluster=1 1454 557 224 64
Cluster=2 2682 1363 475 98
Cluster=3 6958 2849 919 173
Cluster=4 7926 5436 2356 296
Time (days)

Table 3. Time-Dependent AUC and C-index (Testing set)

	PCA		FAMD	
	AUC	C-index	AUC	C-index
Short-term (1-year)	0.772	0.759	0.771	0.704
Intermediate-term (3-year)	0.777	0.755	0.777	0.707
Long-term (5-year)	0.783	0.753	0.783	0.712

Conclusion

- PCA-driven clustering identified distinct patient phenotypes that significantly improved survival prediction beyond baseline demographics (age and sex).
- FAMD-based clustering identified patient phenotypes that significantly improved survival prediction beyond age and sex alone.
- Unsupervised clustering via both PCA and FAMD captured clinically meaningful patterns in the EHR and ECG data. Even though the clusters were not created using MACE outcomes, they were strongly associated with future cardiovascular risk.

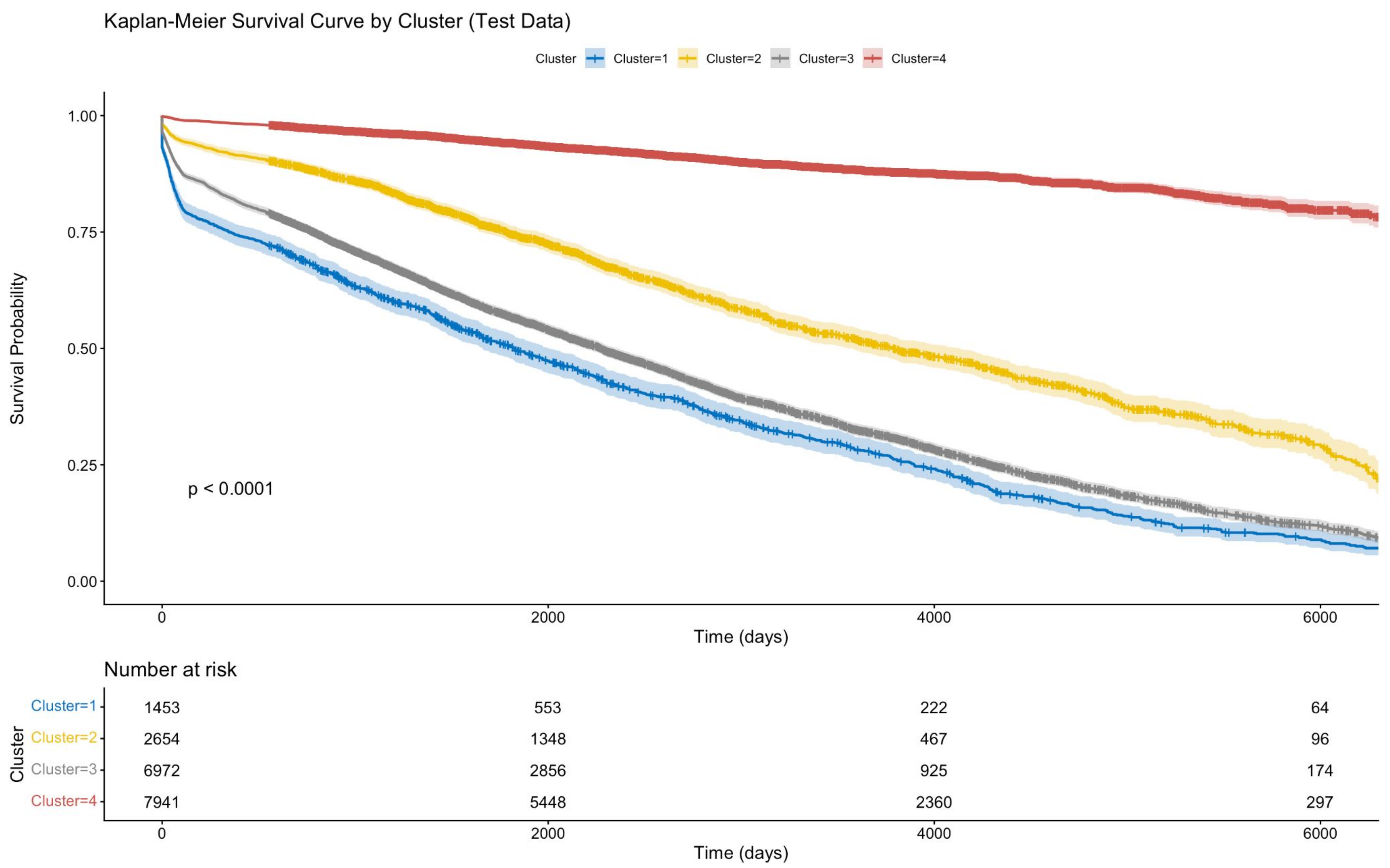
Discussion

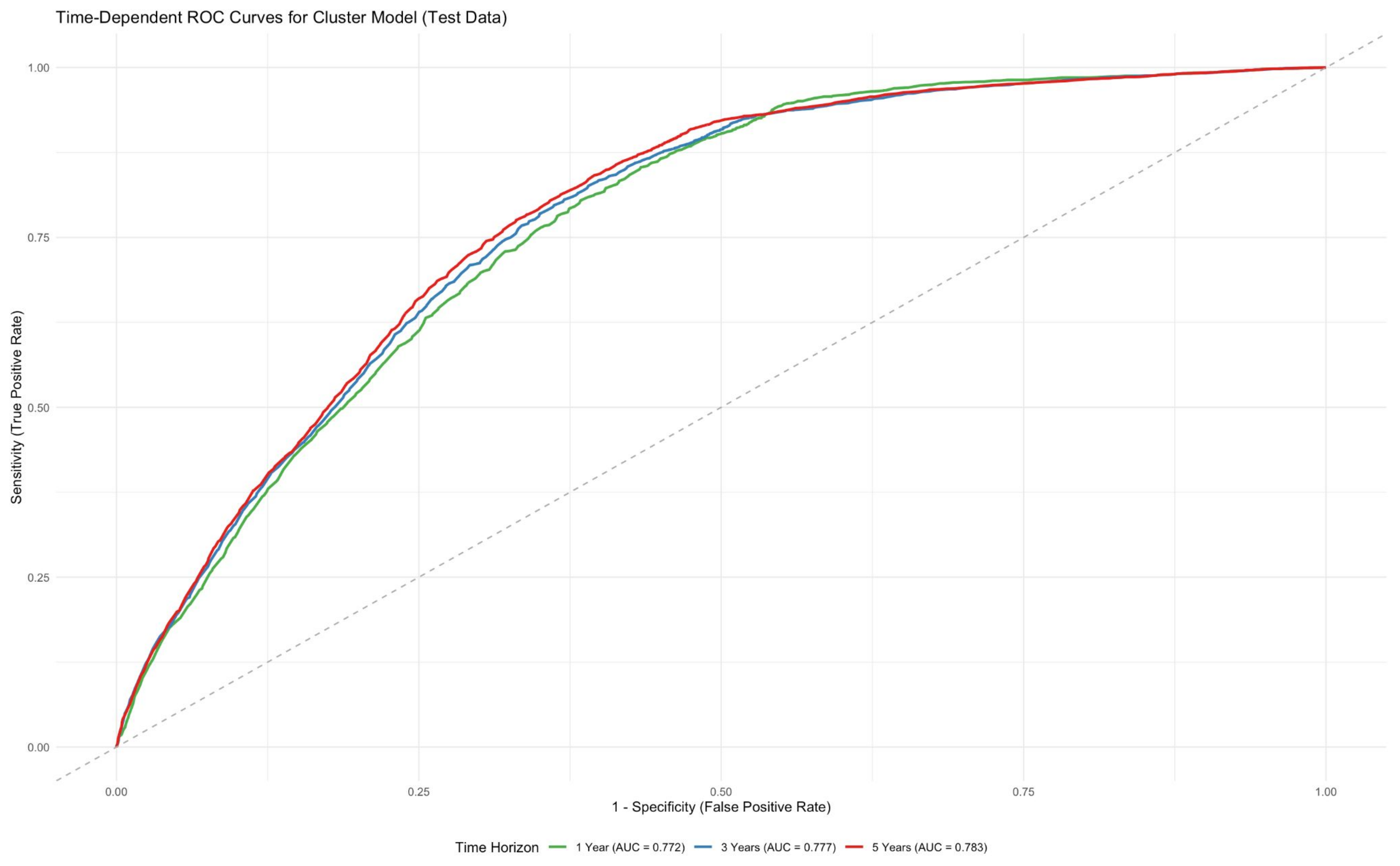
- **Unsupervised phenotyping:** Both PCA and FAMD identified clinically meaningful T2D subgroups from baseline EHR and ECG features before using MACE outcomes.
- **Risk stratification:** Both PCA- and FAMD-derived clusters showed different MACE risks; Cluster 1 had older patients and higher MACE rates, while Cluster 4 had younger patients and lower MACE rates.
- **Model improvement:** Adding cluster membership to the Cox model improved prediction, suggesting clusters provide useful prognostic information.
- **PCA–FAMD comparability:** FAMD was aligned with the PCA workflow using the same imputed data, 80/20 split, k-means clustering, and Cox/time ROC evaluation.
- **Limitation:** Dimensionality reduction was performed before the train-test split; a stricter validation approach would fit FAMD only on the training data and project the test data afterward.

Reference

1. González-Juanatey, C., Anguita-Sánchez, M., Barrios, V., Núñez-Gil, I., Gómez-Doblas, J. J., García-Moll, X., Lafuente-Gormaz, C., Rollán-Gómez, M. J., Peral-Disdier, V., Martínez-Dolz, L., Rodríguez-Santamarta, M., Viñolas-Prat, X., Soriano-Colomé, T., Muñoz-Aguilera, R., Plaza, I., Curcio-Ruigómez, A., Orts-Soler, E., Segovia, J., Fanjul, V., Cequier, Á., … Savana Research Group (2022). Major Adverse Cardiovascular Events in Coronary Type 2 Diabetic Patients: Identification of Associated Factors Using Electronic Health Records and Natural Language Processing. *Journal of clinical medicine*, *11*(20), 6004. https://doi.org/10.3390/jcm11206004
2. American Diabetes Association. *Diabetes Care*. 2023;46(Suppl. 1):S19-S40.
3. Ahlqvist, E., Storm, P., Käräjämäki, A., Martinell, M., Dorkhan, M., Carlsson, A., Vikman, P., Prasad, R. B., Aly, D. M., Almgren, P., Wessman, Y., Shaat, N., Spégel, P., Mulder, H., Lindholm, E., Melander, O., Hansson, O., Malmqvist, U., Lernmark, Å, . . . Groop, L. (2018). Novel subgroups of adult-onset diabetes and their association with outcomes: a data-driven cluster analysis of six variables. *The Lancet Diabetes & Endocrinology, 6*(5), 361–369. 10.1016/S2213-8587(18)30051-2
4. McCoy, R. G., Swarna, K. S., Deng, Y., Herrin, J. S., Ross, J. S., Kent, D. M., Borah, B. J., Crown, W. H., Montori, V. M., Umpierrez, G. E., Galindo, R. J., Brito, J. P., Mickelson, M. M., & Polley, E. C. (2024). Derivation of an Annualized Claims-Based Major Adverse Cardiovascular Event Estimator in Type 2 Diabetes. *JACC. Advances*, *3*(4), 100852. https://doi.org/10.1016/j.jacadv.2024.100852
5. Bresso, E., Lacomblez, C., Duarte, K., Monzo, L., Baudry, G., Tromp, J., Sharma, A., & Girerd, N. (2025). Unsupervised machine learning for cardiovascular disease: A framework for future studies. *European journal of heart failure*, *27*(12), 3366–3377. https://doi.org/10.1002/ejhf.70076

PCA + K-Means Results


Kaplan-Meier Survival Curve by Cluster (Test Data)
Cluster Cluster=1 Cluster=2 Cluster=3 Cluster=4
Survival Probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
Time (days)
Number at risk
Cluster
Cluster=1 1453 553 222 64
Cluster=2 2654 1348 467 96
Cluster=3 6972 2856 925 174
Cluster=4 7941 5448 2360 297
0
2000
4000
6000
Time (days)


Time-Dependent ROC Curves for Cluster Model (Test Data)
Sensitivity (True Positive Rate)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1 - Specificity (False Positive Rate)
Time Horizon 1 Year (AUC = 0.772) 3 Years (AUC = 0.777) 5 Years (AUC = 0.783)

```
Likelihood ratio test

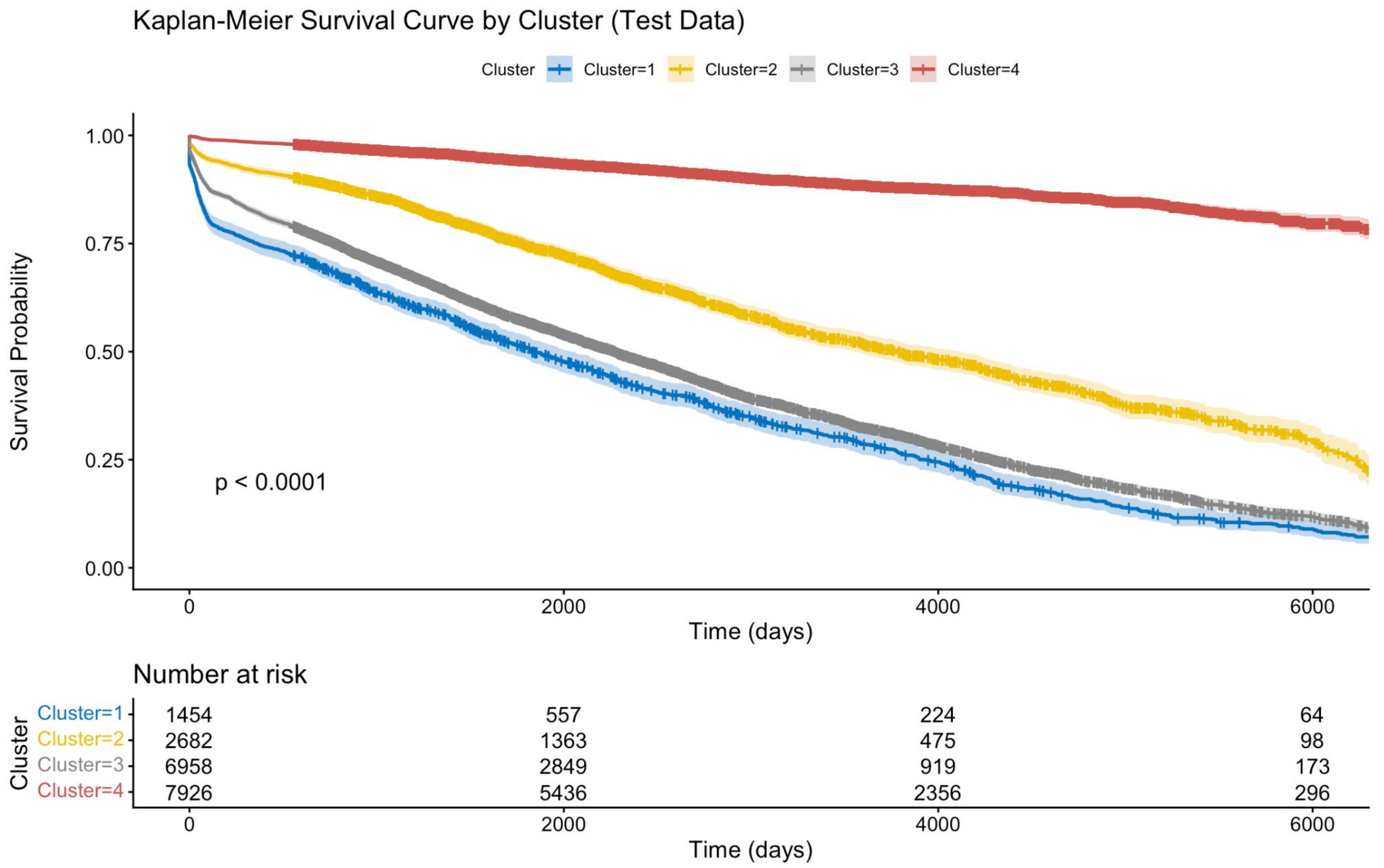
Model 1: Surv(follow_up_days, comp_outcome) ~ demographics_age_index_visit +
    demographics_birth_sex
Model 2: Surv(follow_up_days, comp_outcome) ~ demographics_age_index_visit +
    demographics_birth_sex + Cluster
  #Df  LogLik Df Chisq Pr(>Chisq)
1   2 -315973
2   5 -310111  3 11725  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

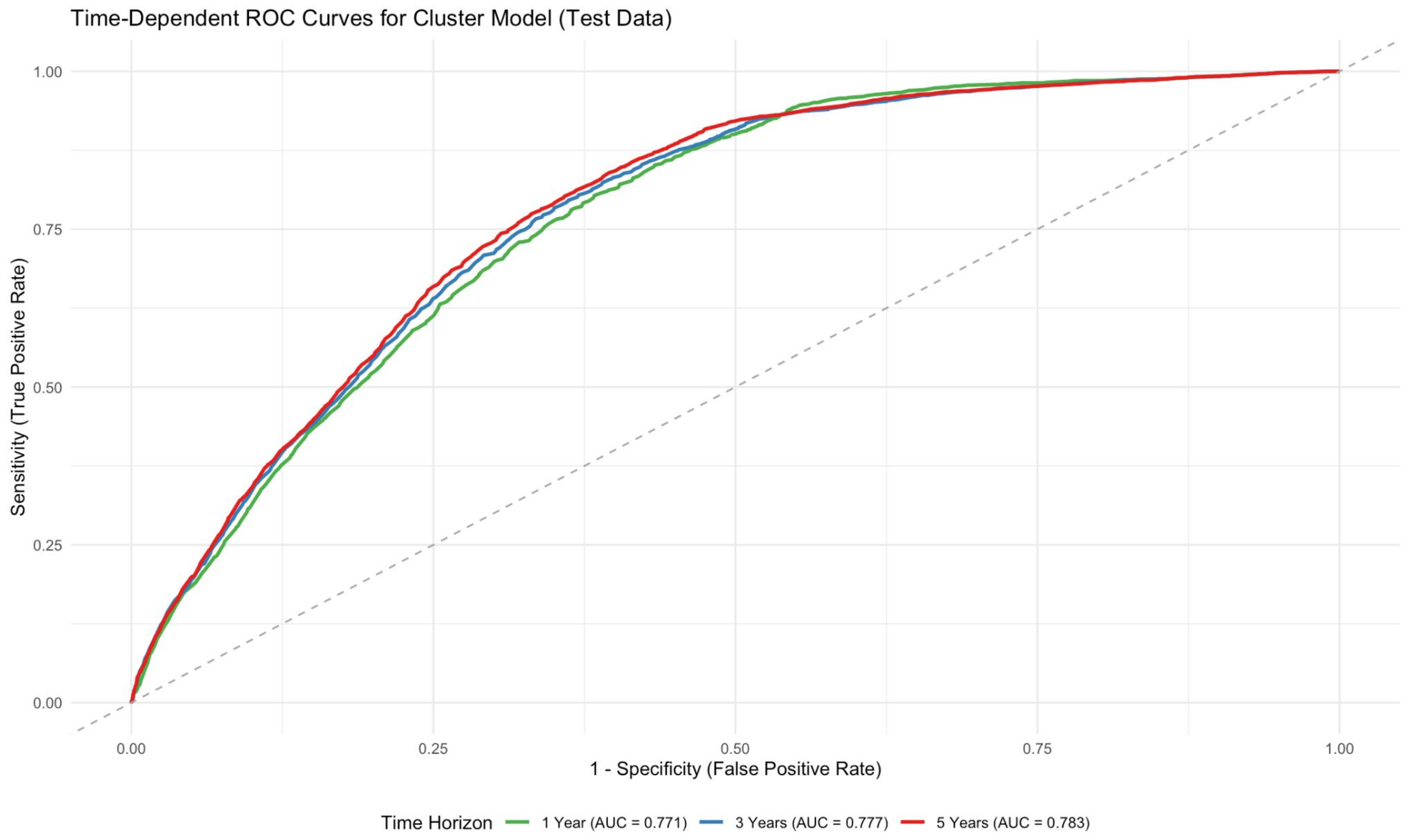
	Horizon	Evaluable N	Events	AUC	C-index
t=365	1 year	19,020	1,964	0.772	0.759
t=1095	3 year	17,546	3,325	0.777	0.755
t=1825	5 year	15,486	4,606	0.783	0.753

Table 1: Cluster Profile (Training Set)

Cluster	N	Percent	Mean age	Female %	Male %	Overall MACE	1y MACE	3y MACE	5y MACE
1	5,976	7.9%	72.4	32.1%	67.9%	79.6%	26.0%	38.0%	47.5%
2	10,228	13.4%	64.0	11.4%	88.6%	43.0%	7.7%	14.5%	22.3%
3	27,625	36.3%	68.1	44.2%	55.8%	65.7%	16.9%	28.6%	39.3%
4	32,251	42.4%	53.0	48.7%	51.3%	10.9%	1.9%	3.8%	5.8%

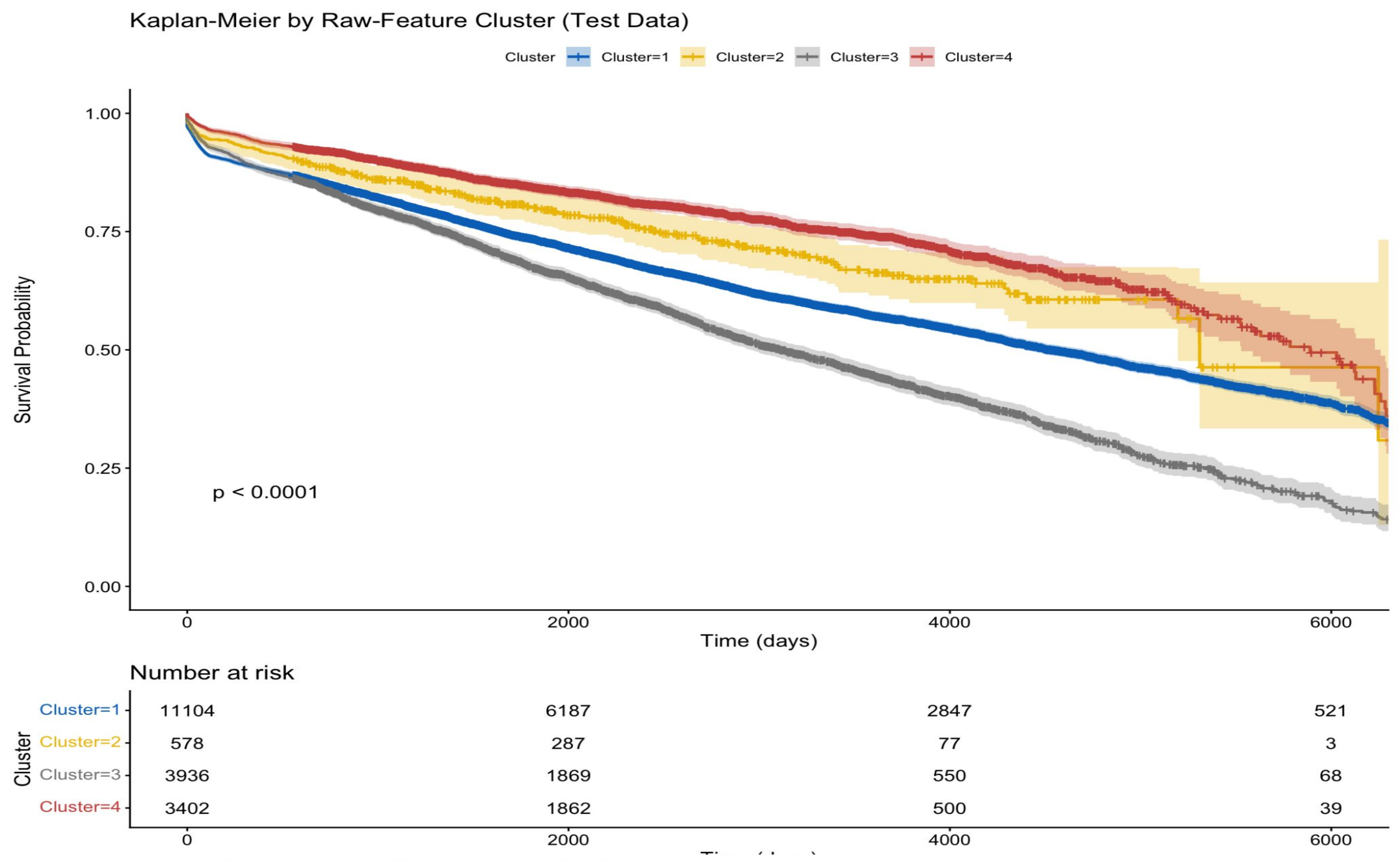
FAMD + K-Means Results

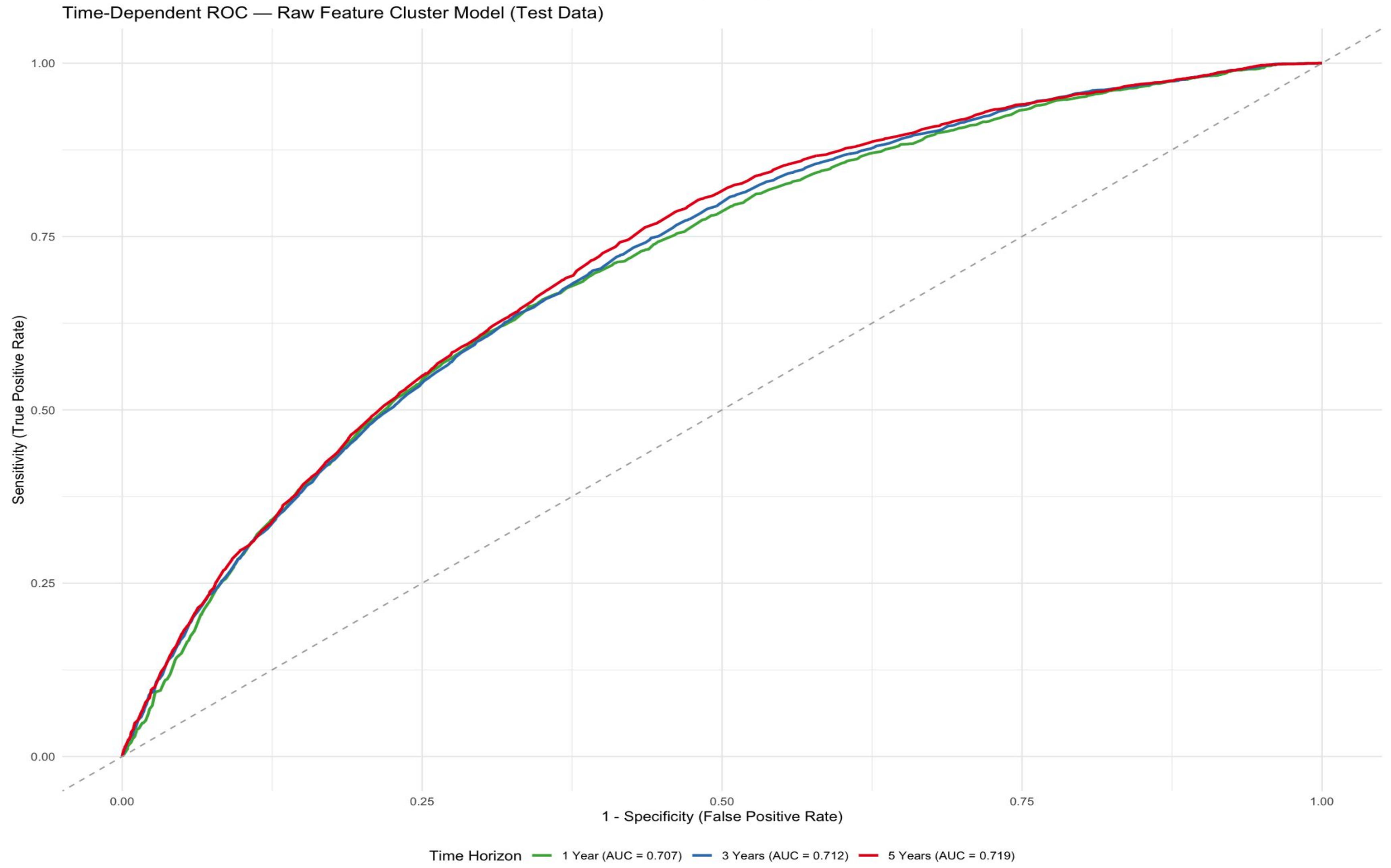

Kaplan-Meier Survival Curve by Cluster (Test Data)
Cluster Cluster=1 Cluster=2 Cluster=3 Cluster=4
Survival Probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
Time (days)
Number at risk
Cluster
Cluster=1 1454 557 224 64
Cluster=2 2682 1363 475 98
Cluster=3 6958 2849 919 173
Cluster=4 7926 5436 2356 296
0
2000
4000
6000
Time (days)


Time-Dependent ROC Curves for Cluster Model (Test Data)
Sensitivity (True Positive Rate)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1 - Specificity (False Positive Rate)
Time Horizon 1 Year (AUC = 0.771) 3 Years (AUC = 0.777) 5 Years (AUC = 0.783)

N	Percent	Mean age	Female %	Male %	Overall MACE	1y MACE	3y MACE
5,982	7.9%	72.4	32.0%	68.0%	79.3%	25.9%	38.3%
10,340	13.6%	64.0	11.5%	88.5%	43.2%	7.8%	14.9%
27,526	36.2%	68.1	44.3%	55.7%	65.8%	20.0%	29.2%
32,232	42.4%	52.8	48.6%	51.4%	10.9%	1.9%	3.9%

Raw data + K-Means Results


Kaplan-Meier by Raw-Feature Cluster (Test Data)
Cluster Cluster=1 Cluster=2 Cluster=3 Cluster=4
Survival Probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
2000
4000
6000
Time (days)
Number at risk
Cluster
Cluster=1 11104 6187 2847 521
Cluster=2 578 287 77 3
Cluster=3 3936 1869 550 68
Cluster=4 3402 1862 500 39
0
2000
4000
6000


Time-Dependent ROC — Raw Feature Cluster Model (Test Data)
Sensitivity (True Positive Rate)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1 - Specificity (False Positive Rate)
Time Horizon 1 Year (AUC = 0.707) 3 Years (AUC = 0.712) 5 Years (AUC = 0.719)

Cluster	N	Percent	Mean age	Female %	Male %	Overall MACE	1y MACE	3y MACE
1	55,737	58.6%	61.7	0.0%	59.3%	43.5%	11.1%	18.2%
2	2,867	3.0%	59.9	0.0%	51.0%	26.6%	8.5%	14.3%
3	19,616	20.6%	63.3	0.0%	63.6%	50.5%	11.1%	20.2%
4	16,880	17.7%	58.8	0.0%	56.0%	22.0%	5.6%	10.1%